

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.3 ANIMAL GASEOUS EXCHANGE AND RESPIRATION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Think back to the very first lesson, when you measured your own breathing rate and pulse rate at rest and again after 2 minutes of step-ups. Your driving question is: Why does my body work so much harder to get oxygen into my cells during exercise, and what happens inside those cells when there is not enough? Write your answers in the space under each prompt. Use your own Lesson 1 data, your final cumulative model and your Driving Question Board. Write in full sentences and use the scientific words you have learned. Explain the reasons (the 'why' and 'how'), not only what happened. Where a prompt asks you to apply your learning to a new situation, use the same ideas.

Part 1: What Your Body Data Showed

In Lesson 1 you measured your breathing rate and pulse rate at rest and then straight after 2 minutes of vigorous step-ups.

(a) Describe what happened to both measurements across the class, using example numbers from the class data table. (b) Which two body systems were responding together? (c) Write down the one observation from that day that your early ideas could not explain.

(a) In every learner both rates rose sharply. For example, a learner who breathed 14 times per minute at rest breathed about 28 times per minute after the step-ups, so breathing rate roughly doubled. A resting pulse of 72 beats per minute rose to over 120 beats per minute. (b) The breathing (respiratory) system and the circulatory system (heart and blood) were responding at the same time. (c) Some learners felt a burning sensation in their thigh muscles even though they were breathing hard the whole time. Our early ideas could not explain why muscles burn when oxygen is being breathed in.

Part 2: Getting Oxygen from the Air into the Blood

Explain how oxygen gets from the air outside your body into your blood during exercise. In your answer: (a) explain how the diaphragm and the intercostal muscles make air move into and out of the lungs,

(a) Breathing is a pressure-driven process. During inhalation the diaphragm contracts and flattens and the external intercostal muscles raise the rib cage. This increases the volume of the thorax, so the pressure inside falls below atmospheric pressure and air flows in. During exhalation the muscles relax, thoracic volume decreases, pressure rises and air is pushed out. After the step-ups these muscles contract more often and more forcefully, so more air, and therefore more oxygen, reaches the lungs each minute. (b) The four features are: a large surface area (millions of alveoli, so many gas molecules can diffuse at once); thin walls, one cell thick (a short diffusion distance); a moist surface (oxygen dissolves before crossing); and a rich blood supply (it carries oxygen away and keeps a steep concentration gradient). (c) A tilapia also needs to take oxygen in quickly and has to do so from

and how this links to your doubled breathing rate; (b) name the four features that make the alveoli an efficient gas exchange surface, and explain what each one does for diffusion; (c) explain why tilapia gills in Lake Victoria need the same four features.	water. Its gill lamellae have a large surface area, thin walls, a moist surface and many capillaries, so oxygen can diffuse into the blood fast enough, just as in alveoli.
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Part 3: Transport and the Heart–Lung Link

Trace the journey of one oxygen molecule from an alveolus to a working muscle cell in your leg, naming each structure or process in order. Then explain why your heart rate rose at the same time as your breathing rate, and describe what happens to carbon dioxide on its way out.	An oxygen molecule diffuses from the alveolus across the thin alveolar wall and capillary wall into the blood, because oxygen is at a higher concentration in the alveolus than in the blood. It binds to haemoglobin in a red blood cell. The blood travels in the pulmonary vein to the left side of the heart, which pumps it into the aorta and on to the muscle capillaries. There the oxygen leaves the haemoglobin and diffuses into the muscle cell, moving from high to low concentration. The heart rate rose because the heart is the pump that delivers oxygen to the cells. Beating faster increases the speed of oxygen delivery to the muscles and the speed at which carbon dioxide is removed. The lungs raise the supply of oxygen and the heart raises how fast it is delivered, so both must work harder. Carbon dioxide travels the reverse route: it diffuses from the muscle cells into the capillaries, is carried by the blood to the lungs, diffuses into the alveoli and is breathed out.
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Part 4: What Happens Inside the Cells

(a) Write the word equation and the balanced chemical equation for aerobic respiration, say where in the cell it happens, and explain how it accounts for the doubling of your breathing rate during step-ups. (b) Explain why your leg muscles burned even though you were breathing hard. Use the terms anaerobic respiration, lactic acid and oxygen debt. (c) Explain why you still breathe heavily for a while after you stop exercising.	(a) Word equation: glucose + oxygen → carbon dioxide + water + ATP. Balanced equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$. It happens in the mitochondria. Contracting muscles need much more ATP during step-ups, so aerobic respiration speeds up. It uses more glucose (from food such as ugali) and more oxygen and produces more carbon dioxide. The body therefore has to breathe faster to supply oxygen and remove carbon dioxide. (b) When oxygen delivery to the muscle cells cannot keep up with demand, aerobic respiration alone cannot supply enough ATP. The cells switch to anaerobic respiration, which releases energy without oxygen but gives far less ATP. In animal muscle this produces lactic acid. The acid builds up in the muscle and causes the burning sensation, even though the lungs are working hard. (c) During intense exercise an oxygen debt builds up. Heavy breathing continues afterwards to supply the extra oxygen needed to break down the accumulated lactic acid into carbon dioxide and water.
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Part 5: Transfer – Akinyi Swims the Winam Gulf

Akinyi, a competitive swimmer from Kisumu, attempts to cross the Winam Gulf of Lake Victoria. Predict what happens to her breathing rate and pulse rate as she swims hard, and explain why in terms of her muscle cells. Then explain what might cause her muscles to ache if she swims very fast, and what her body does when she gets out of the water. Finally, give one reason why her body temperature would rise.

As Akinyi swims hard, her breathing rate and pulse rate will both rise. Her muscle cells need much more ATP, so aerobic respiration in their mitochondria speeds up and uses more oxygen and glucose and makes more carbon dioxide. Her diaphragm and intercostal muscles work faster to bring in more air, oxygen diffuses across her alveoli into the blood and is carried by haemoglobin, and her heart beats faster to deliver it to her muscles and carry carbon dioxide away. If she swims very fast, oxygen delivery cannot meet demand, so her muscle cells use anaerobic respiration and make lactic acid, which causes aching and burning in her muscles. When she gets out of the water she keeps breathing heavily to repay the oxygen debt by breaking down the lactic acid into carbon dioxide and water. Her temperature rises because cellular respiration releases heat as a by-product, and her enzymes work best at about 37 °C, so she sweats and more blood flows to her skin to keep her cool.

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the class data	Accurately uses the Lesson 1 numbers (e.g. 14 to 28 breaths per minute, 72 to over 120 beats per minute) and links them to the explanation throughout.	Mentions that both rates rose, with some correct numbers, but rarely links them to the later explanation.	Gives vague or incorrect data, or no data.
Explaining gas exchange and transport	Explains breathing mechanics with pressure changes, the four alveolar features and the full oxygen pathway including haemoglobin and the heart, with correct diffusion reasoning.	Describes most of the mechanics, features and pathway but with a missing step or limited reasoning.	Lists some structures with little explanation, or has errors in the pathway.
Explaining cellular respiration and lactic acid	Gives both equations correctly, places respiration in the mitochondria, and explains lactic acid, burning legs and oxygen debt as cause and effect.	Gives the equations and a mostly correct explanation of lactic acid with minor gaps.	Equations are incomplete or incorrect, or the burning sensation is not explained.
Transfer to the Akinyi scenario	Applies the whole chain of ideas to the new situation, including oxygen debt and heat, using correct vocabulary.	Applies most ideas to the swimming scenario with some omissions.	Repeats facts without applying them to Akinyi.
Scientific vocabulary and clarity	Uses unit terms accurately and writes in clear, connected sentences.	Uses most terms correctly with generally clear writing.	Uses few scientific terms or uses them incorrectly.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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